

KC MIT
Complete Education

2025

"Group A"

Attempt all the Questions:

1. Circle(O) the correct answer.
 - i) Every node of a Singly Linked List contains fields.
 - a) One
 - ☒ b) Two
 - c) Three
 - d) No
 - ii) Which of the following is not a hash function?
 - a) Division remainder
 - b) Folding
 - c) Chaining
 - d) Mid square
 - iii) How does the rear index increment in circular queue?
 - ☒ a) $\text{front} = (\text{rear} + 1) \% \text{SIZE}$
 - b) $\text{rear} = (\text{rear} + 1) \% \text{SIZE}$
 - c) $\text{rear} = \text{rear} + 1$
 - d) $\text{rear} = (\text{rear} - 1) \% \text{SIZE}$
 - iv) What is the complexity of linear search?
 - ☒ a) $O(n)$
 - b) $O(n \log n)$
 - c) $O(1)$
 - d) $O(\log n)$
 - v) What is the recursive formula for in order traversal of binary tree?
 - ☒ a) Left-Root-Right (LVR)
 - b) Root-Left-Right (VLR)
 - c) Left-Right-Root (LRV)
 - d) Right-Left-Root (RLV)
 - vi) What is the number of disk movement in TOH with 4 disks?
 - a) 9
 - ☒ b) 15
 - c) 7
 - d) 8
 - vii) Which of the following is not related with a leaf node?
 - a) Terminal node
 - b) Degree 0
 - c) With no children
 - ☒ d) With more than one child
 - viii) Which of the following operator has the highest precedence?
 - a) *
 - ☒ b) ^
 - c) +
 - d) -
 - ix) A node except the root node of a B-Tree should be at least full.
 - a) Two-third
 - ☒ b) Half
 - c) 75%
 - d) All a, b and c
 - x) From which end an element is dequeued from a queue?
 - a) top
 - b) rear
 - ☒ c) front
 - d) middle

"Group B"

Attempt any SIX questions

[6 * 5 = 30]

2. Define Data Structure. Explain various types of Data structure. [1+4]
3. What is stack? Explain the push and pop operations on stack using linked list. [1+4]
4. What is the advantage of using doubly linked list over singly linked list?
5. Sort the below data using selection sort algorithm: 1, 15, 17, 10, 7, 21, 5, 20, 13. [4]
6. Draw a Binary Search Tree from the String TRIBHUVAN and traverse the tree in preorder, inorder and postorder. [2+3]
7. Write function or algorithm to search an element in Binary Search Tree. [5]
8. What do you mean by Transitive closure of graph? Find transitive closure of the relation $R = \{(1,1), (1,3), (2,2), (2,4), (3,3), (4,2), (4,3)\}$ on set $A = \{1,2,3,4\}$.

"Group C"

Attempt any TWO questions

[2 * 10 = 20]

9. Define Minimum Spanning Tree (MST). Write Kruskal's algorithm to solve MST problem. Also find the weight of MST by drawing a graph. [1+4+5]
10. Define Hash function. What is hash collision? State different techniques to resolve hash collision. Insert the following elements into a Hash table of size 1000 using: (a) Division and (b) Mid-Square function: 1122, 1234, 5421, 3131, 123 [1+1+2+6]
11. Define a Binary Tree. Differentiate between Strictly binary tree and complete binary tree along with their diagram. Explain deletion of node from binary search tree along with the algorithm. [1+3+6]

Kantipur College of Management and Information Technology
Preliminary Examination

KCMIT
Complete Education

Full Marks: 60
Time: 3 hrs

2025

BCA / Third Semester / CACS 204: Object Oriented Programming in JAVA

Multiple choice questions

Group A

(10*1=10)

1. Which of the following is not a class in java?
a. ArrayList b. String c. BufferedWriter ☒ d. Finalize
2. Which of the following is not a modifier in java?
a. public b. private c. protected ☒ d. synchronized
3. What is the return type of the run() method in the Object class?
a. int b. void c. long d. String
4. Which package contains the Scanner class?
☒ a. java.util b. java.io c. java.swing d. java.lang
5. Which of the following is not a method of String class?
a. hashCode() b. concat() c. trim() d. charAt()
6. In java, JDK stands for _____.
a. java development kit
b. java deployment kit
c. java debugging kit
d. java data kit
7. Which of the following is not a class in java?
a. ArrayList b. String c. BufferedWriter ☒ d. Finalize
8. Which of the following is not component in swing
a. JTextField b. JButton c. JMenu d. JPasswordField
9. Which option is false about the final keyword?
☒ 1. A final method cannot be overridden in its subclasses.
☒ 2. A final class cannot be extended.
☒ 3. A final class cannot extend other classes.
☒ 4. A final method can be inherited.
10. In which memory a String is stored, when we create a string using new operator?
☒ a. Stack b. String memory c. Heap memory d. Random storage space

Attempt any six question

Group B

(5x6=30)

- 1) Define Constructor. Explain the types of constructor along with example. [5]
- 2) Why method overloading is called early binding while overriding is called dynamic binding? Illustrate with syntax. [5]
- 3) Write a java program to read string from user and write in a file named "abc.txt" and open that file again to read and display in console using BufferedReader and BufferedWriter. [5]
- 4) Write a program to create one child thread extending thread to print "hello java" in console also sleep thread for 1000 millisecond. [5]
- 5) Explain all the layout managers in java? [5]
- 6) Write a program to take five colors as string from user and store them using object of ArrayList and check whether color blue is present. [5]
- 7) Write a program to take two string from user and do the following: [1+2+2]
 - i. Check if both the strings are same.
 - ii. Delete the leading and trailing space of first string.

Comprehensive Question

Group C

(2x10=20)

- 8) Differentiate between checked and unchecked exception. Write a java program to handle ArrayIndexOutOfBoundsException? [10]
- 9) Write a java program to do the following:
 - a) To design UI for the login module as shown in the figure. [4]

Username	
Password	
LOGIN	REGISTER

- b) You must write a code to handle button click event, so that when user clicks a "REGISTER" button you must save user name and password to file and when user click "LOGIN" button you must read saved user name and password to validate against user input user name and password. [6]

All the Best

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Preliminary Examination

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2025

Full Marks: 60
Time: 3 hrs

BCA / Thrd Semester / CACS 204: System Analysis and Design

Group A

Attempt all questions

[1X10=10]

1. In the implementation phase of system analysis and design following are included.....
 - a. Parallel run
 - b. sizing
 - c. specification freeze
 - d. all of the above
2. modifying the software to match changes in ever changing environment is called?
 - a. Adaptive maintenance
 - b. Corrective maintenance
 - c. Preventive maintenance
 - d. Perfective maintenance
3. The data flow diagram shows
 - a. The flow of data
 - b. process
 - c. area where they are stored
 - d. all of them
4. At highest level DFD is referred to as ...
 - a. Scope diagram
 - b. Context diagram
 - c. Level 1 DFD
 - d. Level 2 DFD.
5. The shortest time in which a project can be completed best describes
 - a. slack
 - b. critical path
 - c. Expected path
 - d. Optimistic path
6. In mathematical term table is referred as ?
 - a. Relation
 - b. Attribute
 - c. Tuple
 - d. Domain
7. In mathematical term row is referred as?
 - a. relation
 - b. Attribute
 - c. Tuple
 - d. Domain

8. The first phase of sdlc is?
 - a. planning
 - b. Analysis
 - c. design
 - d. maintenance
9. which of the following is viewed as top down source?
 - a. Business unit
 - b. Steering committee
 - c. Managers
 - d. Development groups
10. Feasibility factors include?
 - a. Political
 - b. legal
 - c. technical
 - d. All of above

Group B

Attempts any six Questions.

[5X6=30]

1. Describe the project identification and selection process.
2. What are the three generic competitive strategies.
3. Explain the types of information system.
4. Explain the prototyping approach of SDLC in details. brief.
5. Describe five methods of interacting with the system.
6. Give the real world factors affecting maintenance cost.
7. Differentiate user documentation and system documentation.

Group C

Attempt any two questions.

[2X10=20]

1. What is contained in BPP? Are the content and format of all baseline plan are same?
2. Discuss the reason why organization undertake information system project.
3. What is business process? Why is business process diagramming important?

2025

BCA / Thrd Semester / CACS 205 : Web Technology

Candidates are required to answer all the questions in their own words as far as practicable

Group B

Short Answers Questions:

[5 × 6 = 30]

1. Create XML file that represents your family tree. Assumptions can be made where required.
2. How display and position properties are used in web pages? Explain with example.
3. What is cookie? How cookie is manipulated? Explain with example.
4. What is XML? Explain XSLT, XPath and XQuery with example.
5. Identify the properties for background formatting in CSS. Explain each of the property with their values.

Group C

Comprehensive Questions:

[2 × 10 = 20]

6. Create signup form and login form and also write program to simulate sign up and login process.
7. Create table of contents (use CSS to generate index for the chapters and contents) as follows:

- 1 Technical overview
 - 1.1 Services offered
 - 1.2 Protocols supported
 - 1.3 Hardware
 - 1.4 Addressing
 - 1.5 GPRS modems and modules
- 2 Coding schemes and speeds
 - 2.1 Multiple access schemes
 - 2.2 Channel encoding
 - 2.3 Multislot Class
 - 2.3.1 Multislot Classes for GPRS/EGPRS
 - 2.3.2 Attributes of a multislot class
- 3 Usability
- 4 History of GPRS

BCA / Third Semester / CACS 205 : Web Technology

Candidates are required to answer all the questions in their own words as far as practicable

Objective Questions:

[10 × 1 = 10]

1. What is the reason of doing client side validation?
 - a. for fast response
 - b. because server can't validate data
 - c. because data is not available in server
 - d. all of the above
2. Which element is not block element?
 - a. H1
 - b. span
 - c. p
 - d. div
3. Which of the following is not the value for display property?
 - a. None
 - b. inline
 - c. Block
 - d. hidden
4. Which of the following statement is not true for XSLT?
 - a. Transform XML documents to other formats
 - b. Design XML documents
 - c. Query XML Documents
 - d. Stylesheet used for XML
5. How id selector is written in CSS?
 - a. .idname
 - b. Sidname
 - c. #idname
 - d. @idname
6. Which of the following way is correct way to assign value to color property?
 - a. #RRGGBB
 - b. rgb(10,10,10)
 - c. colorname
 - d. all of the above
7. What is Cookie?
 - a. Word processor file
 - b. Tabular file
 - c. Comma separated value file
 - d. Page visitor record file
8. Differentiate between and <div> element.

One is block element and another is

 - a. inline
 - b. Both are block elements
 - c. Both are inline elements
 - d. Both are category less elements
9. Which function is used to execute query in PHP?
 - a. mysqli_fetch_array()
 - b. mysqli_num_rows()
 - c. mysqli_query()
 - d. mysqli_connect()
10. Which is the correct number of values in box-shadow property?
 - a. 1
 - b. 2
 - c. 3
 - d. 4

BCA / Third Semester / CAST 202: Probability and Statistics

Group A

Attempt all the questions.

- (10×1=10)
- Which of the following gives the most repeated value in a data set?
(a) Mean
(b) Mode
(c) Median
(d) Range
 - Which of the following divides a distribution into four equal parts?
(a) Median
(b) Percentiles
(c) Deciles
(d) Quartiles
 - Which of the following is the relative measure of dispersion?
(a) Quartile deviation
(b) Mean deviation
(c) Standard deviation
(d) Coefficient of variation
 - What is the range of correlation coefficient of variables?
(a) 0 to 1
(b) -1 to +1
(c) -1 to 0
(d) -1 to ∞
 - In the regression equation: $Y = 6 + 2.2 X$, what is the slope of the regression line?
(a) 6
(b) 2.2
(c) 8.2
(d) -2.2
 - What is the probability of getting a number more than 4 in rolling a die?
(a) $1/2$
(b) $1/6$
(c) $4/6$
(d) $1/3$
 - What type of sampling technique is used for a homogenous population?
(a) Simple random sampling
(b) Cluster sampling
(c) Stratified sampling
(d) Systematic sampling
 - What are the parameters of a Binomial distribution?
(a) n, p
(b) p, q
(c) n, q
(d) n, p, q
 - What are the mean and variance of a Normal distribution?
(a) n, p
(b) μ, σ^2
(c) np, npq
(d) λ, λ
 - What is F value for one-way classification ANOVA which has 5 rows and 20 observations at $\alpha = 0.05$?
(a) 2.266
(b) 2.8951
(c) 2.7109
(d) 4.5

Group B

Attempt any SIX questions.

[6 x 5 = 30]

2. Discuss on population and sample.

3. Calculate the mean income earned by the employees of an IT company given in the following table.

Income (Rs. '00)	25-50	50-75	75-100	100-125	125-150	150-175	175-200
No. of people	21	45	68	100	70	35	10

4. A computer manager needs to know how efficiency of her new computer program depends on the size of incoming data. Efficiency will be measured by the number of processed requests per hour. Applying the program to data set of different size, the following result is recorded.

Data size (Gb)	6	7	7	8	10	10	12	14	15
Processed request	40	55	50	41	17	26	20	18	16

Calculate the correlation coefficient between data size and processed request. Interpret the result.

5. The sales revenue (in lakhs Rs.) and advertising expenditure (in lakhs Rs.) of a business company for 10 years are presented in the following table.

Sales revenue	34	30	40	42	35	36	45	50	46	42
Advertising exp.	16	12	18	20	15	16	24	30	24	25

(a) Fit the regression model to describe the given data and interpret the estimated regression coefficient.

(b) Estimate the sales revenue when the advertising expenditure is Rs. 28 lakhs.

6. The probability that a girl speaks truth is $\frac{3}{5}$ and that a boy speaks is $\frac{5}{8}$. Find the probability that (a) Both speaks truth and (b) at least one speaks truth.

7. An in-charge of a department store has noticed that a customer who is just browsing will buy something is 0.35. If 12 customers visited the store, what is the probability that

(a) At least one browsing customer will buy?

(b) Non browsing customer will buy?

8. Number of files in folders of a computer programmer is normally distributed with average number of files in a folder is 120 with standard deviation of 30. Find the probability that the number of files in a folder

(i) lies between 100 and 140

(ii) more than 110.

Group C

Attempt any TWO questions.

[2 x 10 = 20]

9. The marks secured in mid-term examination by students of two sections of BIM of a college are given below:

Roll No.	1	2	3	4	5	6	7	8
Section A	34	46	32	44	28	30	31	46
Section B	33	49	20	22	36	34	18	30

(a) Calculate average marks for sections A and B.

(b) Calculate standard deviation and coefficient of variation for both sections.

(c) Which section shows consistent marks?

10. A business researcher selected seven different banks at random and conducted a survey of their interest rates on loans to various companies. The companies were classified into three categories, namely: small, medium and large. The interest rates were presented in the following table.

Bank :	1	2	3	4	5	6	7
Small :	12	11.8	12.3	12.2	12	11	11.5

Medium :	11	11	10.7	10.5	10.8	11.2	11.5
Large :	10.2	9.8	9.2	10	10	9.5	10.4

Can you conclude that there is no difference in the average interest rate charged to companies at 0.05 level of significance?

11. A chemical company wishing to study the effect of extraction time on the efficiency of an extraction operation obtained the data as follows.

Extraction time (min)	27	45	41	19	35	39	19	20	22	29
Extraction efficiency (%)	57	64	80	46	62	72	52	55	59	60

- Fit a straight line to the given data and use to predict the extraction efficiency one can expect when the extraction time is 35 minutes.
- Determine the coefficient of determination and interpret it.